5.3.8.3 High-speed external clock generated from a crystal/ceramic resonator

The high-speed external (HSE) clock can be supplied with a 4 to 48 MHz crystal/ceramic resonator oscillator. All the information given in this paragraph are based on design simulation results obtained with typical external components specified in the table below.

In the application, the resonator and the load capacitors have to be placed as close as possible to the oscillator pins, in order to minimize the output distortion and startup stabilization time. Refer to the crystal resonator manufacturer for more details on the resonator characteristics (frequency, package, accuracy).

Table 33. 4-50 MHz HSE oscillator characteristics

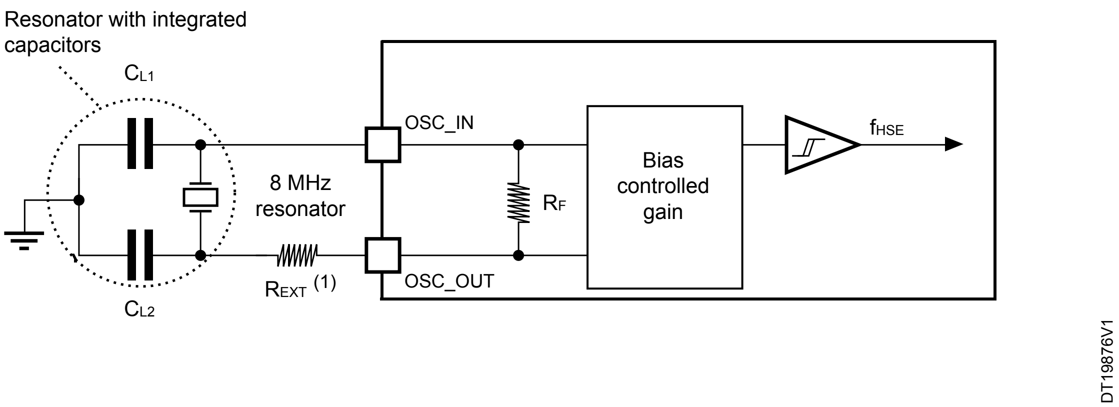
Specified by design and not tested in production.

Symbol	Parameter	Operating conditions(1)	Min	Typ	Max	Unit
F	Oscillator frequency	-	4	-	50	MHz
RF	Feedback resistor	-	-	200	-	kΩ
$I_{DD(HSE)}$	HSE current consumption	During startup(2)	-	-	10	mA
		V_{DD} = 3 V, Rm = 20 Ω C_L=10 pF at 4 MHz	-	0.4	-	
		V_{DD}=3 V, Rm = 20 Ω C_L = 10 pF at 8 MHz	-	0.4	-	
		V_{DD} = 3 V, Rm = 20 Ω CL = 10 pF at 16 MHz	-	0.6	-	
		V_{DD} = 3 V, Rm = 20 Ω C_L = 10 pF at 32 MHz	-	0.7	-	
		V_{DD} = 3 V, Rm = 20 Ω C_L=10 pF at 48 MHz	-	1.2	-	
Gmcritmax	Maximum critical crystal gm	Startup	-	-	1.5	mA/V
t_{SU}(3)	Start-up time	V_{DD} is stabilized	-	2	-	ms

1. *Resonator characteristics given by the crystal/ceramic resonator manufacturer.*
2. *This consumption level occurs during the first 2/3 of the t_{SU}(HSE) startup time.*
3. *$t_{SU(HSE)}$ is the startup time measured from the moment it is enabled (by software) to a stabilized 8 MHz oscillation is reached. This value is measured for a standard crystal resonator and it can vary significantly with the crystal manufacturer.*

Note: For information on selecting the crystal, refer to the application note 'Oscillator design guide for STM8AF/AL/S, STM32 MCUs and MPUs' (AN2867).

Figure 17. Typical application with a 8 MHz crystal



(1): R_{EXT} value depends on the crystal characteristics.